Modern CPU

1. **RISC and CISC Difference**
    - **RISC (Reduced Instruction Set Computing)**:
        - Uses a simplified instruction set designed for efficient hardware performance.
        - Each instruction is executed in a single clock cycle.
        - Smaller and simpler hardware design allows faster processing.
        - Optimized for specific tasks, often used in embedded systems and high-performance servers.
        - Example: PowerPC and G5 processors.
    - **CISC (Complex Instruction Set Computing)**:
        - Employs a wide variety of complex instructions, each capable of executing multiple low-level operations.
        - Requires fewer lines of code in programming but may take longer to execute complex instructions.
        - Typically used in IBM-compatible and general-purpose computers.
        - Example: x86 processors.
2. **Comparison of Processors**
    - **Processing Speed**: Measured in GHz, determines how many cycles per second a CPU can execute.
    - **Cache Size**: Larger cache sizes reduce access time to frequently used data.
    - **Bit Size**: Indicates the width of the processor's registers (e.g., 32-bit vs. 64-bit), affecting the volume of data processed at a time.
    - **Bus Speed**: Determines data transfer rates between the CPU and memory or peripherals.
    - **Manufacturers and Products**:
        - **Intel**: Famous for 4004 (first microprocessor), Pentium, Core i3/i5/i7, and Xeon processors.
        - **AMD**: Competes with Intel, offering products like Athlon, FX, and Opteron series.
        - **IBM**: Developed mainframes and partnered with Apple to create the G5 (first 64-bit consumer processor).
3. **Modern CPU Diagram and Architecture**
    - **Memory**: Stores instructions and data.
    - **Control Unit (CU)**: Manages data flow and directs operations between components.
    - **Arithmetic Logic Unit (ALU)**: Performs arithmetic calculations and logical operations.
    - **Registers**: Small, high-speed storage for temporary data.
    - **Processor Bus**: Connects CPU with memory and peripherals, allowing communication.
4. **ALU**
    - Integral to the CPU, it handles all mathematical calculations (e.g., addition, subtraction) and logical comparisons (e.g., AND, OR operations).
    - Works closely with the Control Unit to execute instructions efficiently.

5. **Microprocessor**
    - **Definition**: A single integrated circuit (IC) containing the core components of a CPU.
    - Known as the **"brain" of the computer** due to its role in managing computations, processing instructions, and controlling hardware operations.
    - Examples: Intel 4004, AMD Opteron, IBM G5.

Storing Data

1. **Advantages, Disadvantages, and Differences in Storage Types**
    - **Magnetic Storage**:
        - **Examples**: Hard drives, floppy disks, tape drives.
        - **Advantages**: Large capacity, cost-effective for large-scale storage.
        - **Disadvantages**: Mechanically vulnerable, slower than newer technologies.
        - **Usage**: Commonly used for desktop/laptop storage and archival data.
    - **Optical Storage**:
        - **Examples**: CDs, DVDs, Blu-rays.
        - **Advantages**: Durable, portable, inexpensive for data distribution.
        - **Disadvantages**: Limited capacity compared to hard drives and SSDs, slower read/write speeds.
        - **Usage**: Ideal for media storage and long-term archival.
    - **Solid-State Storage**:
        - **Examples**: SSDs, USB flash drives.
        - **Advantages**: Fast data access, highly durable, energy-efficient.
        - **Disadvantages**: Expensive and smaller capacity than magnetic storage.
        - **Usage**: Common in portable devices, high-performance systems, and cloud storage.

Computer Software

1. **Definition of Operating System (OS)**
    - The OS is a system software that bridges the gap between hardware and application programs, managing resources like CPU, memory, and input/output devices.
2. **Relation Between Computer and OS**
    - The OS ensures smooth communication between software and hardware, managing operations like file handling, memory allocation, and security protocols.
3. **Functions of an OS**
    - **Resource Management**: Allocates CPU time, memory, and storage.
    - **Task Management**: Prioritizes and schedules multiple tasks for execution.
    - **Memory Management**: Ensures optimal use of memory to avoid conflicts.

    - **File Management**: Provides methods for storing, retrieving, and organizing files.
    - **Security**: Safeguards the system from unauthorized access and malware.
4. **Single-User Multi-Tasking**
    - Supports one user running multiple programs simultaneously.
    - Examples: Windows, MacOS.
5. **What is a Security System?**
    - A combination of software tools designed to protect the computer from threats such as viruses, malware, and unauthorized access.
6. **Symptoms of Virus Attacks**
    - Slow system performance, frequent crashes, missing files, duplicate files, or unusual error messages.

Computer Network

1. **Network Components**
    - **Devices**: Computers, routers, switches.
    - **Connections**: Ethernet cables, fiber optics, or Wi-Fi.
    - **Protocols**: TCP/IP, HTTP, DNS ensure standard communication.
2. **Types of Networks**
    - **LAN**: Covers small areas like homes or offices; faster and secure.
    - **MAN**: Connects LANs within cities or metropolitan areas.
    - **WAN**: Spans large geographic areas, often connecting multiple LANs.
3. **Network Terms**
    - **Host**: A device connected to a network.
    - **Workstation**: A dedicated computer for specific tasks.
4. **Network Architecture**
    - **Peer-to-Peer**: All devices share resources equally without a central server.
    - **Client-Server**: Centralized server provides resources and services to clients.
5. **Data Flow**
    - **Simplex**: Data flows in one direction only.
    - **Half-Duplex**: Two-way communication, but only one at a time.
    - **Full-Duplex**: Two-way communication simultaneously.

E-commerce

1. **Features of E-commerce**
    - 24/7 availability, global reach, and reduced operational costs.
2. **Types of E-commerce**
    - **B2B**: Transactions between businesses (e.g., wholesale suppliers).
    - **B2C**: Businesses selling directly to customers (e.g., Amazon).
    - **B2G**: Businesses providing goods/services to the government.
3. **Digital Marketing and SEO**

    - **SEO**: Enhances visibility by improving website ranking in search engines.
    - **Social Media Marketing**: Builds customer engagement and brand awareness through platforms like Facebook and Instagram.
4. **Significance of Staying Updated**
    - Embracing emerging trends ensures relevance in the competitive e-commerce landscape, such as AI-driven customer support and blockchain for secure transactions.